



Open Source Model Rocket: DESIGN AND SIMULATION

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Rocket design is one of the most exciting yet complex areas of engineering. It offers a completely different perspective on how physics behaves in the real world, where theory meets real forces, real materials, and real-world constraints interacting simultaneously.

In rocketry, careful attention must be given to the centre of gravity (CG) and centre of pressure

(CP), both of which directly influence flight stability. Multiple subsystems, including stabilisation electronics, stage ignition electronics, and flight computation systems, must also operate reliably under extreme conditions. Alongside these systems, the rocket is constantly influenced by gravity, air pressure, aerodynamic drag, surface roughness, material thickness, weight

distribution, and structural strength. Together, these factors define how the rocket performs during flight, making rocketry both technically demanding and deeply engaging.

For most beginners, rocketry begins with model rockets, where the fundamentals of physics, electronics, and fuel ignition chemistry come together in a practical and highly visual way. In many countries, rocketry has evolved into an active ecosystem supported by strong communities, competitions, and hobby-level innovation. However, in India, there is still a noticeable gap in awareness and accessibility within this field. This series aims to document the complete journey of designing a model rocket system while making the learning process more approachable for enthusiasts and beginners alike.

The series covers the entire development process, from initial design concepts and simulations to testing, causes of failures, and iterative improvements. One of the defining aspects of the system is that it will remain open source, allowing readers to freely clone, modify, and build upon the design for personal learning or commercial applications.

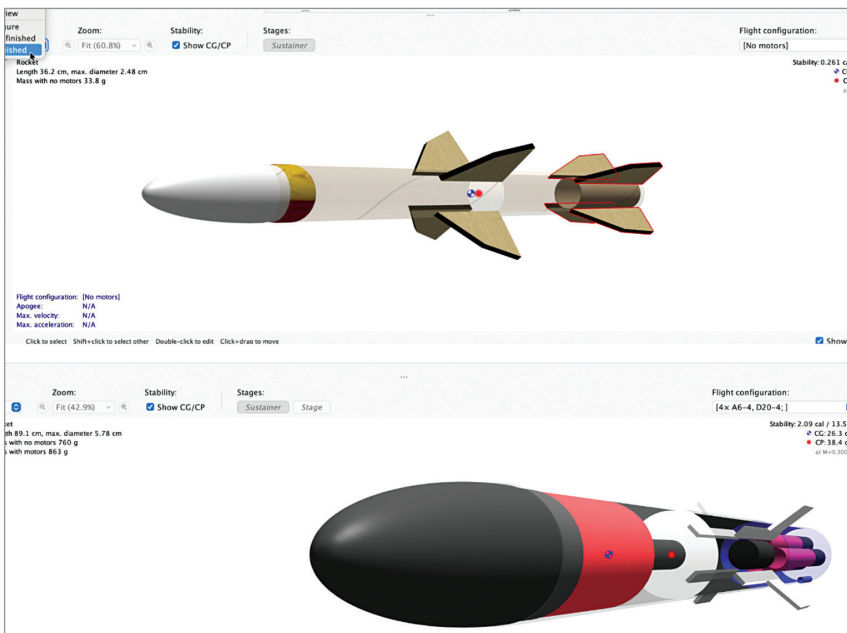


Fig. 1: Model rocket design simulator

TABLE 1
ROCKET BODY

Category	Component	Specification	Recommended Range/Notes
Structure	Body tube	3D-printed body tube or standard market-available part	Diameter and length as per the selected design specifications and part number
Structure	Nose cone (dome)	3D-printed (PLA/PETG) or standard online-store available part	Specifications should match the selected design and best-fit part number obtained during simulation
Structure	Fins	Three fins	Fin dimensions and shape as per simulation and stability requirements
Structure	Model rocket engine	Estes C6-5/ES1507 rocket engine or custom-designed rocket engine	Select the rocket engine according to the model rocket design requirements
Structure	Launch lug	Small tube	Inner diameter: 5mm–8mm, mounted in proper alignment with the rocket body
Structure	Nose weight	Steel or lead weight	Adjusted to shift the centre of gravity (CG) forward for better stability